

Aim:

- Handling missing values: detection, filling, and dropping
- Removing duplicates and unnecessary data
- Data type conversion and ensuring consistency
- Normalize data (e.g., standardization, min-max scaling).

CODE:

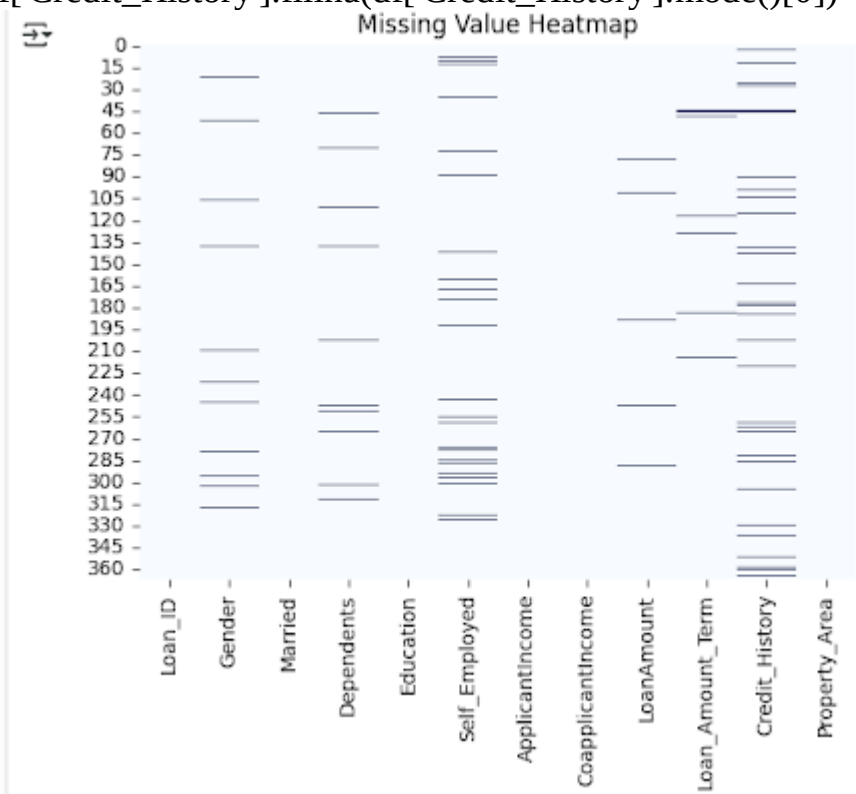
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')
print("Initial Data Overview:")
print(df.info())
```

```
Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Loan_ID               367 non-null   object
 1   Gender                356 non-null   object
 2   Married               367 non-null   object
 3   Dependents            357 non-null   object
 4   Education             367 non-null   object
 5   Self_Employed         344 non-null   object
 6   ApplicantIncome       367 non-null   int64
 7   CoapplicantIncome     367 non-null   int64
 8   LoanAmount            362 non-null   float64
 9   Loan_Amount_Term      361 non-null   float64
10   Credit_History         338 non-null   float64
11   Property_Area         367 non-null   object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None
```

```
print("\nMissing Values in Each Column:\n", df.isnull().sum())
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
```

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```
plt.title("Missing Value Heatmap")
plt.show()
for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
    df[col].fillna(df[col].mode()[0])
df['LoanAmount'].fillna(df['LoanAmount'].median())
df['Loan_Amount_Term'].fillna(df['Loan_Amount_Term'].mode()[0])
df['Credit_History'].fillna(df['Credit_History'].mode()[0])
```



Credit_History	
0	1.0
1	1.0
2	1.0
3	1.0
4	1.0
...	...
362	1.0
363	1.0
364	1.0
365	1.0
366	1.0

367 rows x 1 columns

dtype: float64

```
initial_rows = df.shape[0]
```

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```
df.drop_duplicates(inplace=True)
print(f"\nRemoved {initial_rows - df.shape[0]} duplicate rows.")
```



```
Removed 0 duplicate rows.
```

[6]

```
df['Dependents'] = df['Dependents'].replace('3+', 3).fillna(0).astype(int)
for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
    df[col] = df[col].str.strip().str.capitalize()
min_max_scaler = MinMaxScaler()
scale_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])
scaler = StandardScaler()
df[['Credit_History']] = scaler.fit_transform(df[['Credit_History']])
print("\nCleaned Data Summary:")
print(df.info())
print(df.head())
```



```
Cleaned Data Summary:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                367 non-null   object
1   Gender                 356 non-null   object
2   Married                367 non-null   object
3   Dependents             367 non-null   int64
4   Education              367 non-null   object
5   Self_Employed          344 non-null   object
6   ApplicantIncome         367 non-null   float64
7   CoapplicantIncome       367 non-null   float64
8   LoanAmount              362 non-null   float64
9   Loan_Amount_Term        361 non-null   float64
10  Credit_History          338 non-null   float64
11  Property_Area           367 non-null   object
dtypes: float64(5), int64(1), object(6)
memory usage: 34.5+ KB
None
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001015   Male     Yes         0      Graduate             No
1  LP001022   Male     Yes         1      Graduate             No
2  LP001031   Male     Yes         2      Graduate             No
3  LP001035   Male     Yes         2      Graduate             No
4  LP001051   Male     No         0  Not graduate             No

   ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0      0.078865         0.000000      0.157088             360.0
1      0.042411         0.062500      0.187739             360.0
2      0.068938         0.075000      0.344828             360.0
3      0.032263         0.106083      0.137931             360.0
4      0.045168         0.000000      0.095785             360.0

   Credit_History  Property_Area
0      0.459858      Urban
1      0.459858      Urban
2      0.459858      Urban
3         NaN      Urban
4      0.459858      Urban
```

Activate

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Result:

Thus the EDA-Data Cleaning is done successfully.